CPE301 – SPRING 2024

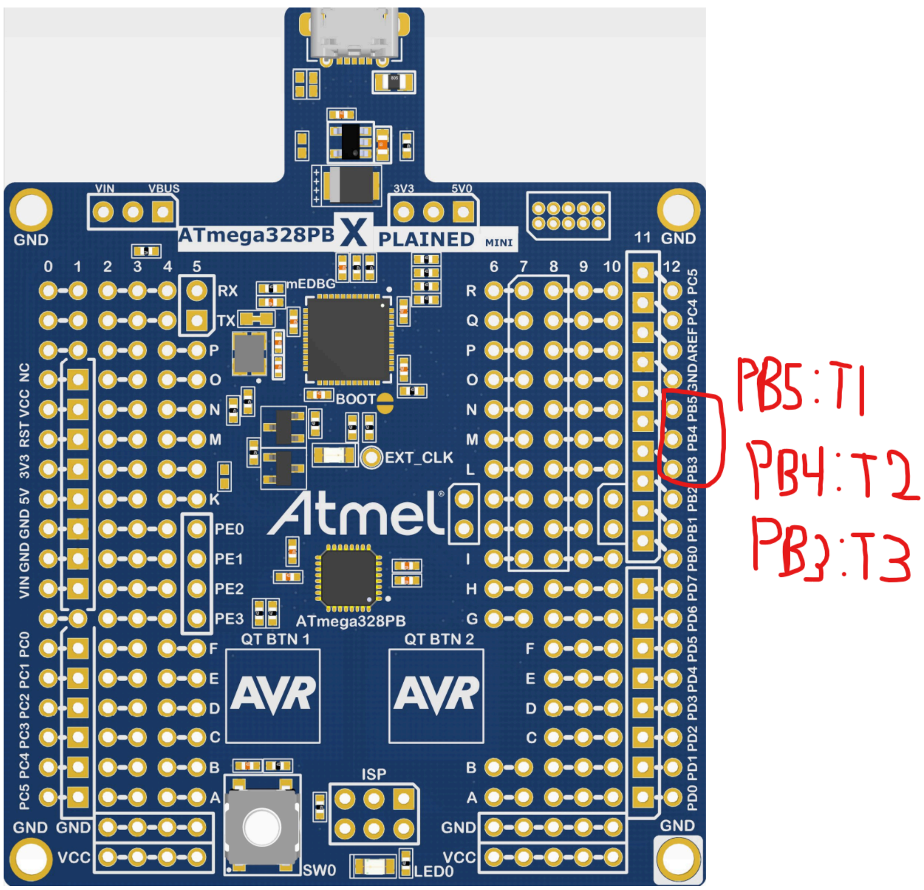
Design Assignment 3

Student Name: Joshua Martinez
Student #: 5007531628
Student Email: marti87@unlv.nevada.edu
Primary Github address: https://github.com/JoshuaMa2003
Directory: https://github.com/JoshuaMa2003/submission_da
Video Playlist:
https://www.youtube.com/playlist?list=PLZ09MA_uFt63me60r5XWtDUj2m21n4nUP

1. COMPONENTS LIST AND DIAGRAM SHOWING PINS USED

List of Components used

- **Microchip Studio Debugger**
- **Microchip Studio Simulator**
- **ATmega328PB Microcontroller**
- **Saleae Logic Analyzer Software**
- **Logic Analyzer**
- **Female-to-Male Wire**



2. INITIAL CODE OF TASK 1

```c
/*
 * CPE301_DesignAssignment3_Task1.c
 *
 * Created: 3/21/2024 7:26:10 PM
 * Author : joshu
 */

#define F_CPU 16000000UL                                //define cpu clock speed at 16Mhz
#include <avr/io.h>

int main(void){
    DDRB |= (1 << 5);                                   //set PB5 as an output

    TCCR0B |= (1 << CS01) | (1 << CS00);                //initialize timer 0 with prescaler 64

    while(1) {                                          //program main loop

        unsigned long overflow = 0;                     //overflow counter that is used for timing

        PORTB |= (1 << 5);                              //turn on PB5 LED

        while(overflow < 977) {                         //loop until overflow counter reaches 977 (approx. 1 second)

            if(TIFR0 & (1 << TOV0)){                    //check if timer 0 has overflowed

                TIFR0 |= (1 << TOV0);                   //clear timer 0 overflow flag

                overflow++;                             //increment overflow counter
            }
        }

        PORTB &= ~(1 << 5);                             //turn off PB5 LED

        overflow = 0;                                   //reset the overflow counter

        while(overflow < 977){                          //timing loop repeated for LED off

            if(TIFR0 & (1 << TOV0)){                    //check if timer 0 has overflowed

            TIFR0 |= (1 << TOV0);                   //clear timer 0 overflow flag

                overflow++;                             //increment overflow counter
            }
        }
    }
    return 0;
}
```

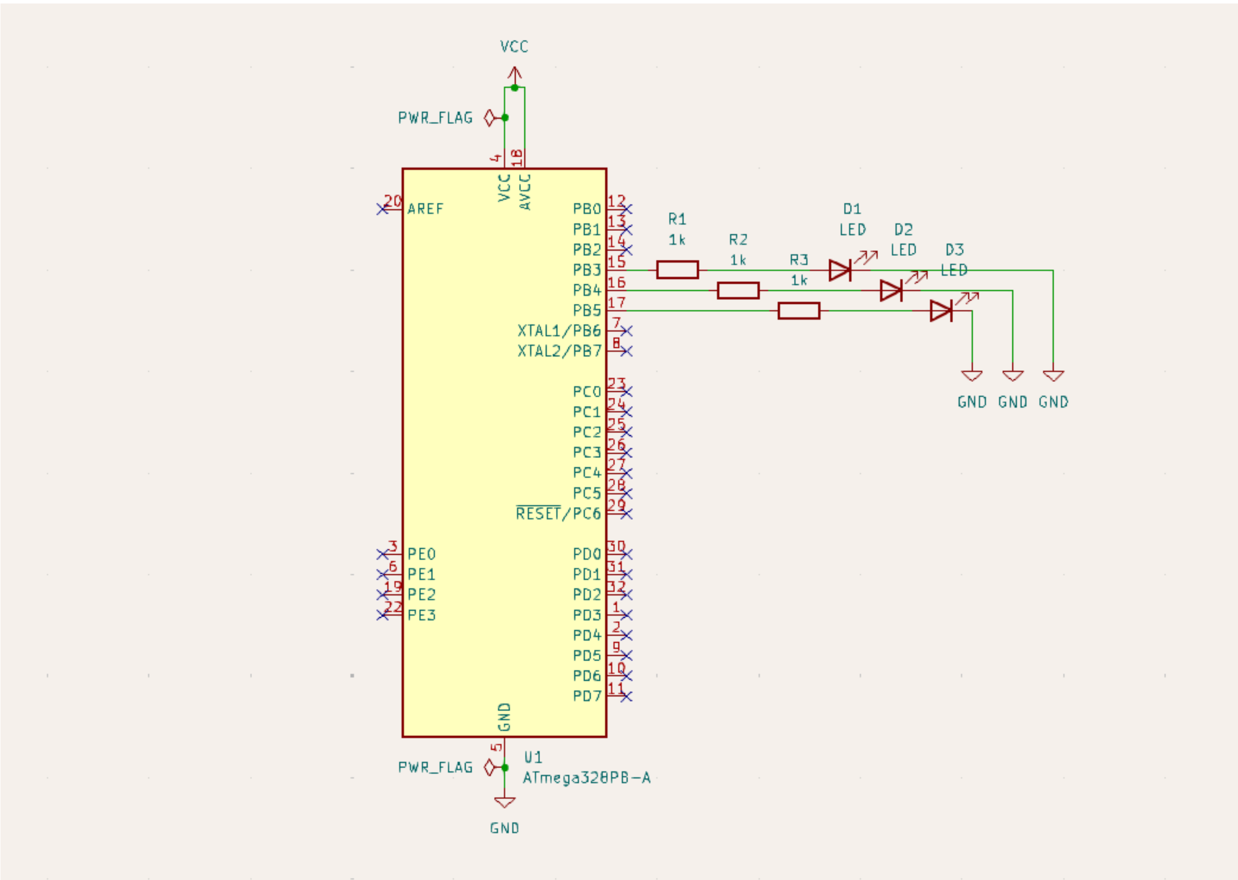
2. INITIAL CODE OF TASK 2

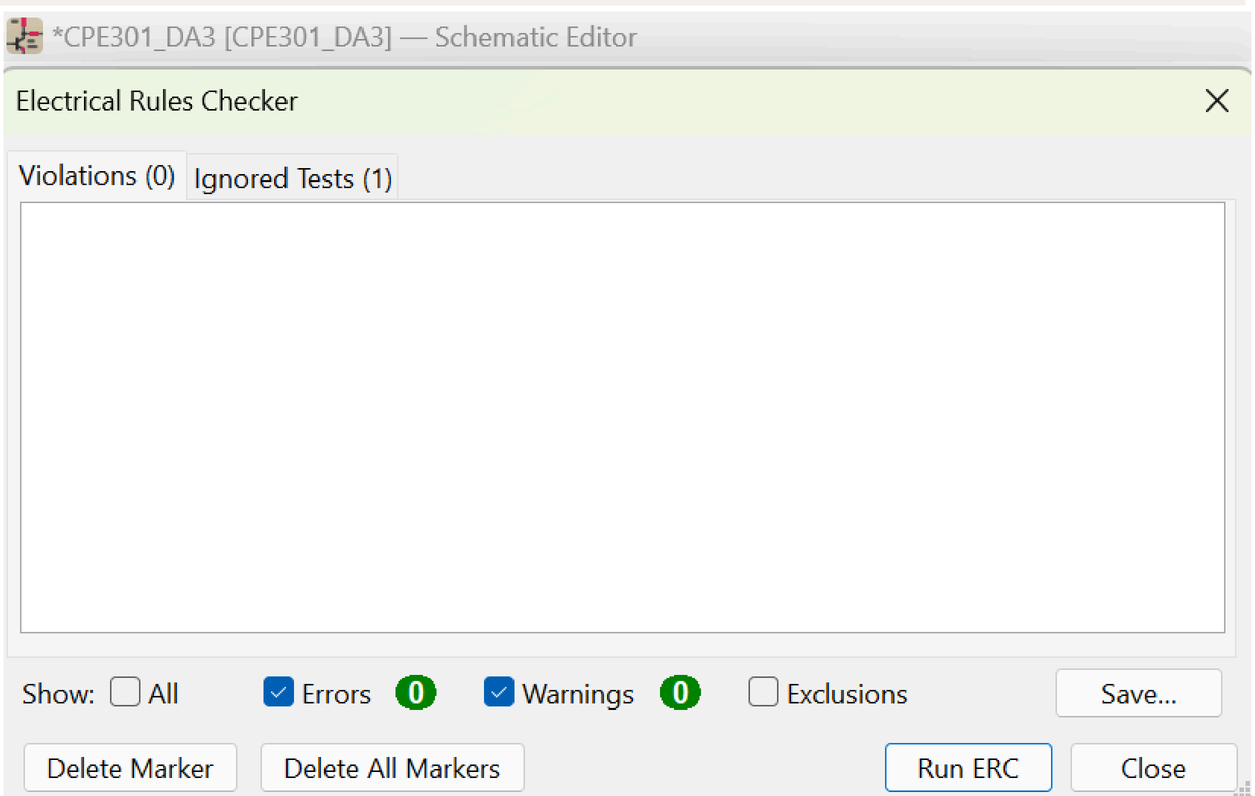
```c
/*
 * CPE301_DesignAssignment3_Task2.c
 *
 * Created: 3/21/2024 7:53:33 PM
 * Author : joshu
 */

#define F_CPU 16000000UL                        //define clock speed at 16Mhz
#include <avr/io.h>
#include <avr/interrupt.h>

volatile unsigned long timer = 0;               //volatile variable to count the number of interrupts


int main(void)
{
    DDRB |= (1 << 4) | (1 << 5);                //set PB4 and PB5 as output pins

    PORTB ^= (1 << 5);                          //initialize LED at PB5 off

    TCCR1B |= (1 << WGM12);                     //Configure timer 1 for CTC mode

    OCR1A = 124;                                //setting Output Compare Register for Timer 1 channel A to 124
                                                //Will cause timer to reset every 0.5 ms ((124 + 1) * prescaler / 16Mhz)
                                                //prescaler in this case would be set to 64

    TIMSK1 |= (1 << OCIE1A);                    //enable Timer 1 Output Compare A Match Interrupt

    TCCR1B |= (1 << CS11) | (1 << CS10);        //setting prescaler to 64

    sei();                                      //enable global interrupts

    while(1){                                   //loop does nothing, waits for interrupts

    }
    return 0;
}

ISR(TIMER1_COMPA_vect){
    timer++;                                    //increment timer variable every time interrupts occurs

    if(timer >= 6000){                          //check if the timer has reached 6000 (approx. 3 seconds)

        PORTB ^= (1 << 4);                      //toggling the LED at PB4

        timer = 0;                              //resetting the timer variable
    }
}
```

3. INITIAL CODE OF TASK 3

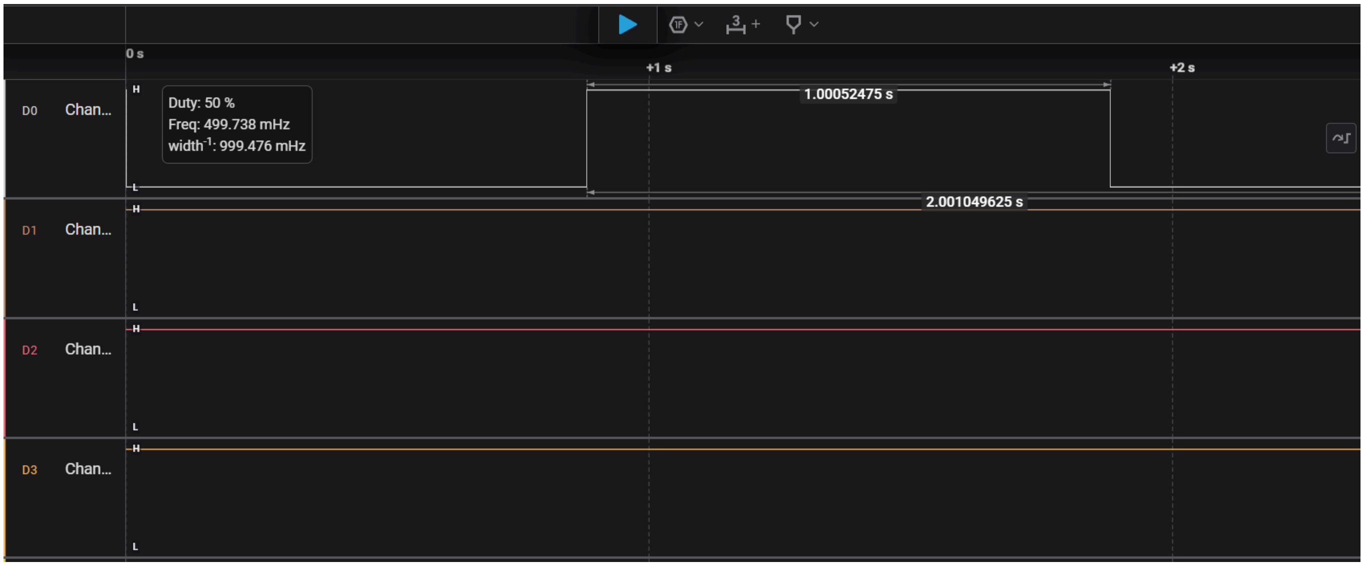
```c
/*
 * CPE301_DesignAssignment3_Task3.c
 *
 * Created: 3/21/2024 10:24:20 PM
 * Author : joshu
 */

#define F_CPU 16000000UL                  //define cpu clock speed at 16Mhz
#include <avr/io.h>
#include <avr/interrupt.h>

volatile unsigned int overflow = 0;         //volatile variable to track number of overflows

int main(void){
    DDRB |= (1 << 3) | (1 << 5);            //setting PB3 and PB5 as outputs

    PORTB ^= (1 << 5);                      //initialize PB5 LED off

    TCCR2B |= (1 << CS22);                  //set prescaler for timer 2 to 64

    TIMSK2 |= (1 << TOIE2);                 //enable timer 2 overflow interrupt

    sei();                                  //enable global interrupts

    while(1){                               //loop does nothing, waits for interrupt

    }
    return 0;
}

ISR(TIMER2_OVF_vect){

    overflow++;                             //increment overflow counter each time timer 2 overflows

    if(overflow >= 1953){                   //check if overflow reaches 1953 (approx. 2 seconds)

        PORTB ^= (1 << 3);                  //toggle the LED at PB3

        overflow = 0;                       //resetting overflow counter
    }
}
```

3. KICAD SCHEMATICS FOR Tasks 1 - 3

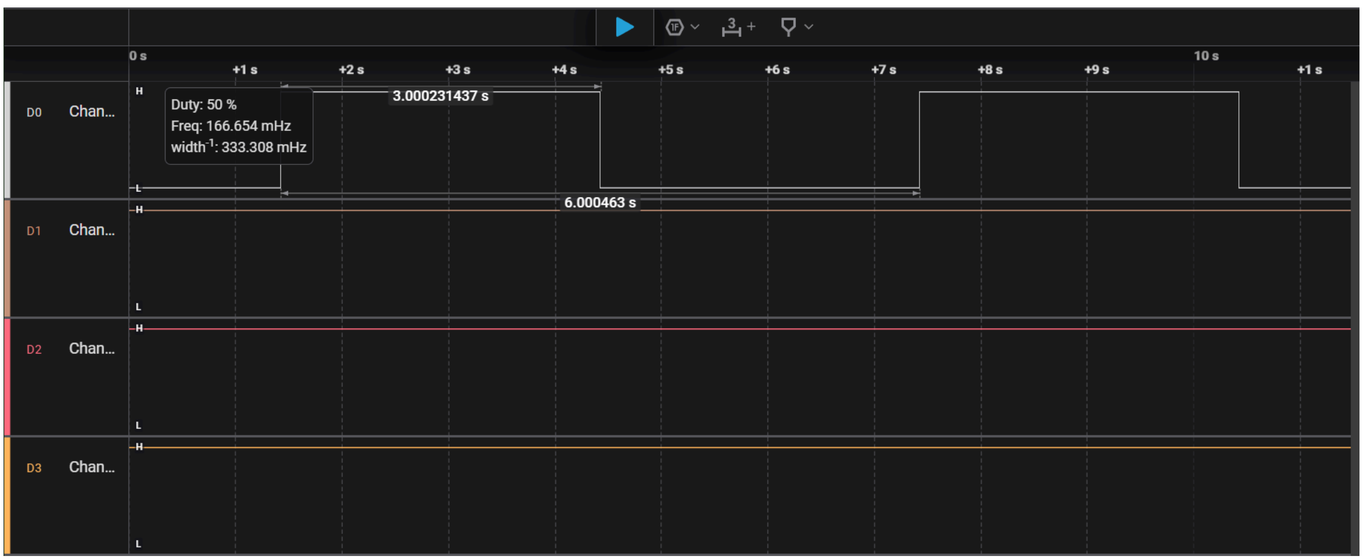




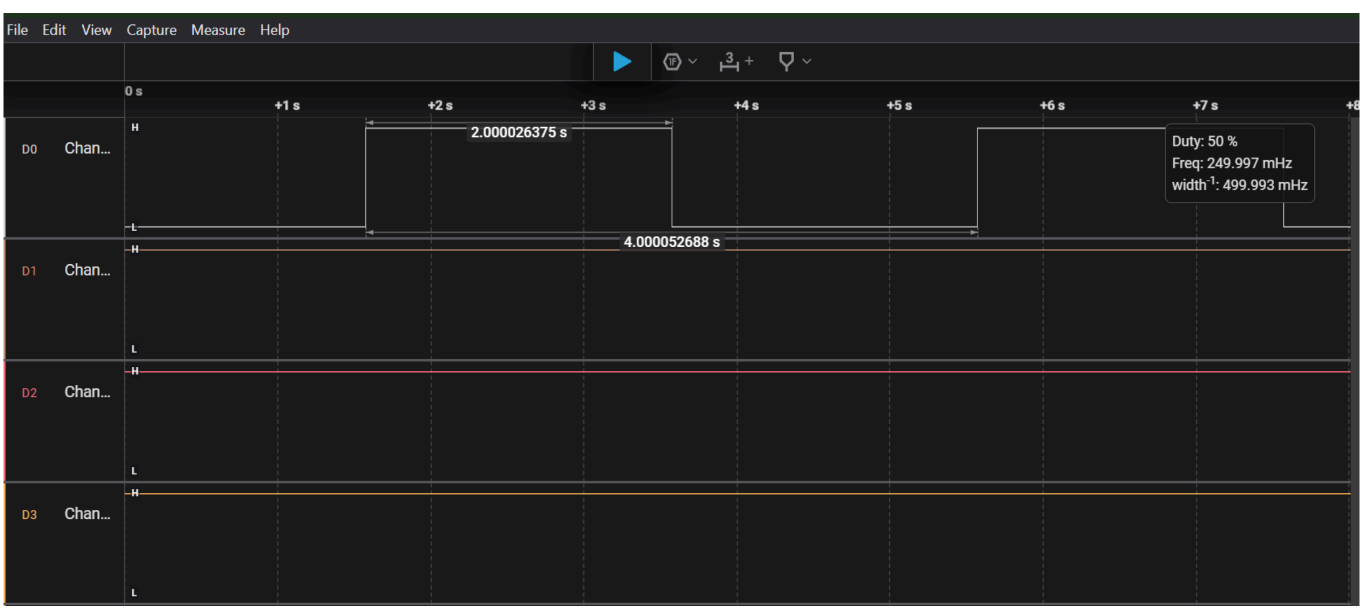
4. SCREENSHOTS OF EACH TASK OUTPUT



Task 1 Output

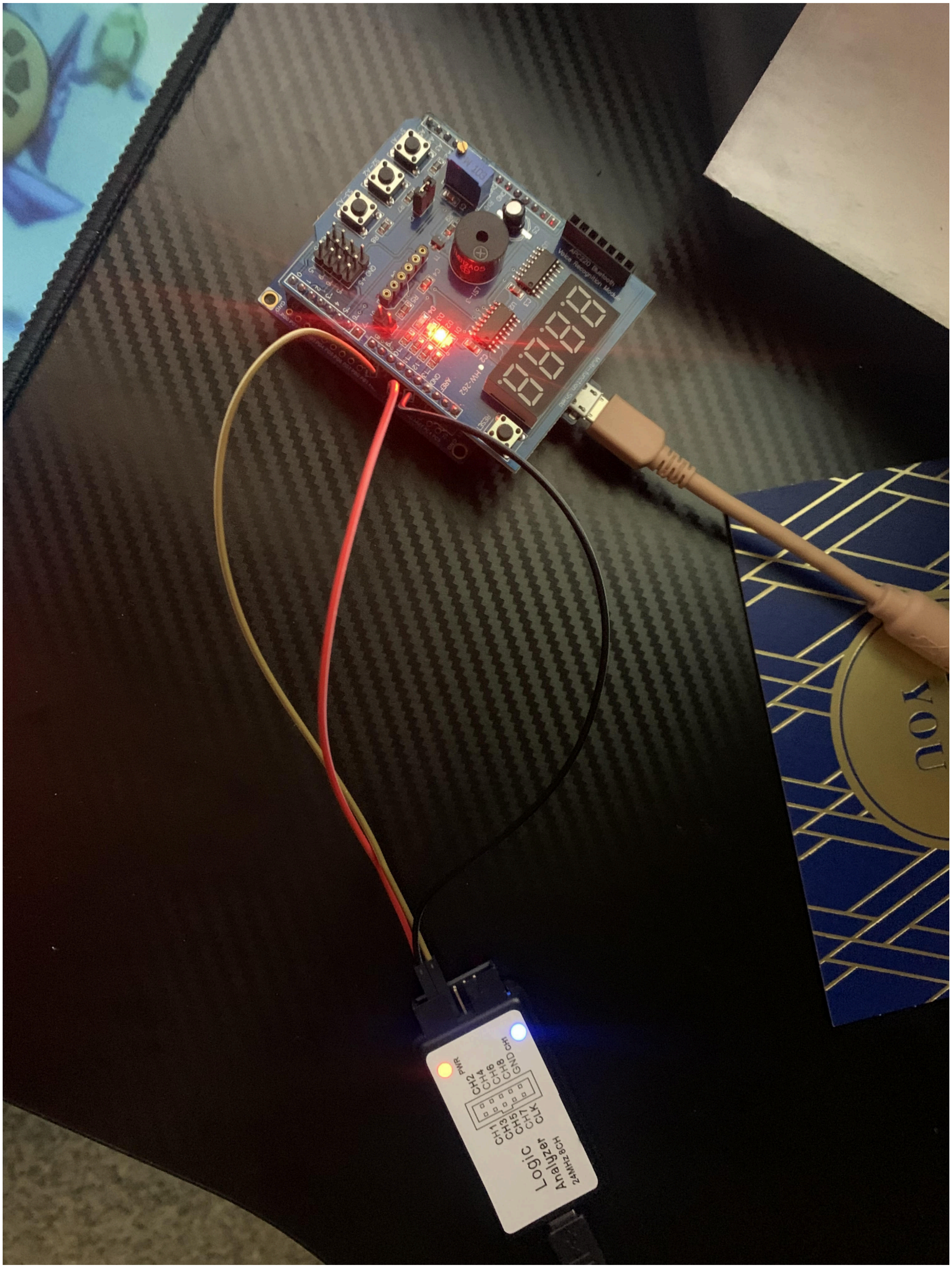


Task 2 Output



Task 3 Output

5. SCREENSHOTS OF EACH DEMO (THE SAME SETUP WAS USED FOR TASKS 1 - 3)



6. VIDEO LINKS OF EACH DEMO

Tasks 1 - 3 Demo: https://youtu.be/6h3iijehkZA?feature=shared

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"This assignment submission is my own, original work".
JOSHUA MARTINEZ